

PETER L. TAYLOR

<https://pltaylor16.github.io/>

Research Focus: Survey cosmology, especially weak lensing, galaxy clustering, statistical inference and machine learning. I am particularly active in DESI and Euclid.

EMPLOYMENT

Research Scientist Center for Cosmology and AstroParticle Physics The Ohio State University	<i>2025 – Present</i>
CCAPP Fellow Center for Cosmology and AstroParticle Physics The Ohio State University (5-year independent fellowship)	<i>2022 – 2025</i>
NASA Postdoctoral Program Fellow Jet Propulsion Laboratory California Institute of Technology (3-year independent fellowship)	<i>2019 – 2022</i>

EDUCATION

PhD, Astrophysics Mullard Space Science Laboratory University College London Thesis: <i>Cosmological Inference with Cosmic Shear</i> Supervisors: Prof. Thomas Kitching, OBE & Prof. Jason McEwen	<i>2016 – 2019</i>
MRes, Astrophysics Durham University Thesis: <i>On the Shape of Dark Matter Halos in the Galaxy Cluster Abell 3827 and the Scattering Cross-Section of Dark Matter</i> Supervisors: Prof. Richard Massey & Prof. Mathilde Jauzac	<i>2015 – 2017</i>
MMATH, Mathematics University of Oxford Dissertation: <i>Kaluza-Klein Cosmologies</i> Supervisor: Prof. Pedro Ferreira	<i>2011 – 2015</i>

PUBLICATION METRICS

54 Papers (**13** First Author, **5** Student First Author); Link to full list
Citations: **> 5000**; h-index: **21**

FUNDING PROFILE

\$950k as PI (or equivalent)
\$1.7 million external funding as named co-I or PI
\$2.3 million total funds as named co-I or PI when including internal JPL R&D grants

AWARDS

Lancelot M. Berkeley–New York Community Trust Prize American Astronomical Society Awarded to the DESI Collaboration	<i>2026</i>
UCL Dean’s Commendation Thesis Prize Faculty of Mathematical and Physical Sciences	<i>2020</i>
Alan Johnstone Award for Outstanding Graduate Research Department of Space and Climate Physics, University College London	<i>2018</i>
UK Science and Technologies Facilities Council Studentship Up to 3.5 Years of Graduate Research Funding	<i>2016</i>

SELECTED GRANTS

Science-PI NASA Astrophysics Theory Program Leveraging Weak Gravitational Lensing - Redshift Space Distortions Cross-correlations (\$748k)	<i>2021</i>
Co-I (1 of 5, PI E Huff) JPL Internal Research and Technology Development Fund Mass and Motion, Tension and Concordance: What Are Tensions in Current Data Telling us About Dark Energy? (\$220k)	<i>2020</i>
Co-I (1 of 1, PI E Huff) JPL Internal Topic Area Proposal Next-Generation Weak Lensing with Hyperspectral Imaging Surveys (\$400k)	<i>2020</i>
Co-I (1 of 10, Science-I B Lee) HST Cycle 28 Archival Study Constraining the masses of galaxy overdensities at $z > 1$ in CANDELS and COSMOS through weak lensing in the NIR (\$751k)	<i>2020</i>
PI NASA Postdoctoral Program Fellowship A Next Generation Statistical Analysis for Next Generation Dark Energy Surveys (~ \$200k)	<i>2019</i>

TEACHING AND MENTORING

Project Supervisor Pavel Demidov PhD Student at New York University	<i>01/26 – Present</i>
Guest Lecturer Astro 5682 – Intro to Cosmology The Ohio State University	<i>10/24</i>
Project Supervisor Alexander Torres Undergraduate at The Ohio State University NSF Graduate Research Fellowship Program (GRFP) (awarded)	<i>06/24 – Present</i>

Project Supervisor Sophie Olsen Undergraduate at The Ohio State University	06/24 – 08/24
Project Supervisor Matthew Craigie PhD Student at University of Queensland Now NPP Fellow at JPL	06/23 – 09/24
Supervisor Erik Zaborowski PhD Student at The Ohio State University NSF Graduate Research Fellowship Program (GRFP) Honorable Mention Ohio State Presidential Fellow Department of Energy Graduate Student Research Award BCCP Fellowship, University of California Berkeley (awarded)	06/22 – Present
Invited Lecturer <i>Euclid Advanced School, Les Houches, France</i> 1.5 hour Lecture on Likelihoods in Cosmology (Video Recording)	06/22
Primary Supervisor Sebastian Tsai Project: <i>The Limits of k-cut 3×2 Point Statistics</i> Caltech Summer Undergraduate Research Fellow & Project Advisor for Senior Thesis at Yale Now Business Analyst at McKinsey	06/21 – 06/22
Primary Supervisor Leah Vazsonyi Project: <i>Constraining $f(R)$ Gravity with k-cut Cosmic Shear</i> Caltech Summer Undergraduate Research Fellow Now PhD student at UNC Chapel Hill	06/20 – 10/21
Project Supervisor Anurag Deshpande PhD student at University College London Now Machine Learning Scientist at Amazon	6/20 – 12/20

TALKS

CCAPP Symposium, The Ohio State University (internal)	09/25
MSSL, University College London (invited)	04/25
University of Edinburgh	03/25
University of Manchester	03/25
Ohio University (invited)	02/25
University of Texas, San Antonio (invited)	02/25
Euclid GC SWG Meeting, Garching, Germany (invited, remote)	01/25

Euclid US Seminar Series (invited, remote)	01/25
University of Chicago (invited)	10/24
University of North Carolina, Chapel Hill	09/24
Duke University	09/24
Key Project 7 Workshop, DESI Conference, Marseille, France (invited, remote)	07/24
Euclid Conference, Rome, Italy	06/24
Cosmology from Home (selected, remote)	06/24
University of Michigan (invited)	06/24
University of Cincinnati (invited)	02/24
Parity Violations from Home 2023 (selected talk, remote, Video Recording)	10/23
CosmoPalooza ¹ (invited, remote, Video Recording)	10/23
CCAPP Symposium, The Ohio State University (internal)	09/23
Lensing on Different Scales Workshop, Chicago (selected talk)	07/23
DESI Meeting, Durham, UK (flash talk)	07/23
Euclid Meeting, Copenhagen (flash talk, selected, remote)	06/23
Statistical Challenges in Modern Astronomy, State College (flash talk)	06/23
Euclid Early Career Talk Series (flash talk, remote)	10/22
CCAPP Symposium, The Ohio State University (internal)	09/22
University of Turin, Italy (invited, remote)	05/22
University of Waterloo, Canada (invited, remote)	02/22
Stanford University (invited, remote)	01/22
Queen Mary University of London (invited, remote)	11/21
Duke University (invited, remote)	10/21
ICG, University of Portsmouth (invited, remote)	10/21
University of California, Santa Cruz (remote)	10/21
Lawrence Berkeley National Lab (remote)	10/21
IPAC, California Institute of Technology (invited, remote)	10/21
University of Geneva (invited, remote)	10/21
USM/LMU, Munich (invited, remote)	09/21
Postdoc Lab-wide Seminar Series, Jet Propulsion Laboratory (remote)	08/21
University of Oxford (invited, remote)	07/21

¹On behalf of the Euclid Consortium

University of Arizona (invited, remote)	03/21
Stanford University (remote)	12/20
Euclid Inter-Science Task Force (IST) Nonlinear Talk Series (invited, remote)	12/20
University of Minnesota (invited, remote)	10/20
External Synergies for Rubin Community Science Workshop ¹ (invited, remote)	08/20
Euclid US Talk Series (remote)	07/20
University of Manchester, Manchester, UK (invited)	08/19
Euclid Science Ground Segment, Euclid Conference, Helsinki, Finland	06/19
Euclid UK Meeting, University of Oxford, Oxford, UK (selected talk)	12/18
Euclid Weak Lensing and Galaxy Clustering Meeting, Milan, Italy	12/18
Alan Johnstone Prize Talk, University College London (internal)	11/18
Euclid France Weak Lensing Atelier, IAP, Paris, France (invited)	10/18
Jet Propulsion Laboratory, California Institute of Technology	08/18
MSSL, University College London (internal)	03/18
ICC/CEA, Durham University (internal)	06/16

PROFESSIONAL ACTIVITIES

Collaboration Membership

Euclid Consortium, Roman Cosmology Science Investigation Team,
Dark Energy Survey, Dark Energy Spectroscopic Instrument,
& Rubin Dark Energy Science Collaboration

Euclid Consortium

Developer, CLOE (Likelihood Pipeline)	2025 - Present
Member, Diversity Committee	2020 - 2023
Co-Lead, Weak Lensing Forward Modelling Work Package	2019 - 2023
Consultant, Likelihood Inter-Science Task-force	2019 - 2024
Science Organizing Committee, Les Houches Advanced School	2022
Internal Referee for Euclid Publications	2023 - Present
Flagship Paper Authorship Rights for > 1 Year of Infrastructure Work	2023 - Present

DESI

In-person Observing	2024
Internal Referee for DESI Publications	2024 - Present
Mentorship Program	2022 - Present

Refereeing and Reviewing

Subject-matter Expert Reviewer in NASA Proposal Peer Review	2021, 2022
Astronomy and Astrophysics	2019 - Present
Monthly Notices of the Royal Astronomical Society	2020 - Present
Journal of Cosmology and Astroparticle Physics	2021 - Present
The Open Journal of Astrophysics	2024 - Present
The Astrophysical Journal	2025 - Present

Organizer

CCAPP Cosmology Journal Club	<i>2025 - Present</i>
CCAPP Seminar Series	<i>2023 - 2024</i>
NASA JPL Dark Sector Meetings	<i>2020 - 2022</i>
Mullard Space Science Laboratory Cosmology Journal Club	<i>2017 - 2018</i>

Fellow, Royal Astronomical Society	<i>2017 - Present</i>
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OUTREACH & PUBLIC ENGAGEMENT

Lead Organizer	<i>07/19</i>
The Universe in Virtual Reality	
Royal Society, London	
Lead Organizer	<i>07/18</i>
Mullard Space Science Laboratory Work Experience Week	
<i>Week long program for high school students from underrepresented backgrounds.</i>	
Project Mentor	<i>07/18</i>
Mullard Space Science Laboratory Work Experience Week	
Outreach Talk	<i>04/18</i>
Institute for the Arts, London	
Project Mentor	<i>07/17</i>
Mullard Space Science Laboratory Work Experience Week	
Public Talk	<i>06/17</i>
Westminster School, London	
Public Demonstrator	<i>05/17</i>
Mullard Space Science Laboratory 50th Anniversary Open Day	
Gravitational Lensing Demonstrator	<i>05/17</i>
Euclid Consortium School Science Day, London	
Demonstrator	<i>03/16</i>
Schools' Science Festival, Durham	
Planetarium Demonstrator	<i>10/15</i>
Celebrate Science Festival, Durham	

FIRST AUTHOR PUBLICATIONS

1. **Peter L. Taylor.** Computing Nonlinear Power Spectra Across Dynamical Dark Energy Model Space with Neural ODEs. *The Open Journal of Astrophysics*, 10.33232/001c.143521, 2025.
2. **Peter L. Taylor**, Andrei Cuceu et al. CombineHarvesterFlow: Joint Probe Analysis Made Easy with Normalizing Flows. *The Open Journal of Astrophysics*, 10.33232/001c.124495, 2024.
3. **Peter L. Taylor**, Matthew Craigie, Yuan-Sen Ting. Unsupervised Searches for Cosmological Parity-Violation: An Investigation with Convolutional Neural Networks. *Phys. Rev. D*, 109:083518, 2024.

4. **Peter L. Taylor** and Katarina Markovič. Covariance of photometric and spectroscopic two-point statistics: Implications for cosmological parameter inference. *Phys. Rev. D*, 106(6):063536, 2022.
5. **Peter L. Taylor**, Katarina Markovič, Alksitis Portsido and Eric Huff. Redshift space distortions: Unmixing radial scales in projection. *Phys. Rev. D*, 105(8):084007, 2022.
- 6.² **Peter L. Taylor** et. al. [94 co-authors]. Euclid: forecasts for k -cut 3x2 point statistics. *The Open Journal of Astrophysics*, 10.21105/astro.2012.04672, 2021.
7. **Peter L. Taylor**, Francis Bernardeau, Eric Huff. x -cut Cosmic Shear: Optimally Removing Sensitivity to Baryonic and Nonlinear Physics with an Application to the Dark Energy Survey Year 1 Shear Data. *Phys. Rev. D*, 103(4):043531, 2021.
8. **Peter L. Taylor**, Thomas D. Kitching, Justing Alsing, Benjamin D. Wandelt, Stephen M. Feeney, and Jason D. McEwen. Cosmic Shear: Inference from Forward Models. *Phys. Rev. D*, 100:023519, 2019.
9. **Peter L. Taylor**, Thomas D. Kitching, and Jason D. McEwen. Nonparametric cosmology with cosmic shear. *Phys. Rev. D*, 99:043532, 2019.
10. **Peter L. Taylor**, Francis Bernardeau, and Thomas D. Kitching. k -cut cosmic shear: Tuneable power spectrum sensitivity to test gravity. *Phys. Rev. D*, 98(8):083514, 2018.
11. **Peter L. Taylor**, Thomas D. Kitching, Jason D. McEwen, and Thomas Tram. Testing the cosmic shear spatially-flat universe approximation with generalized lensing and shear spectra. *Phys. Rev. D*, 98(2):023522, 2018.
12. **Peter L. Taylor**, Thomas D. Kitching, and Jason D. McEwen. Preparing for the cosmic shear data flood: Optimal data extraction and simulation requirements for stage iv dark energy experiments. *Phys. Rev. D*, 98:043532, 2018.
13. **Peter Taylor**, Richard Massey, Mathilde Jauzac, Frederic Courbin, David Harvey, Remy Joseph, and Andrew Robertson. A test for skewed distributions of dark matter, and a possible detection in galaxy cluster abell 3827. *Monthly Notices of the Royal Astronomical Society*, 468(4):50045013, 2017.

PAPERS BY STUDENTS

14. Erik Zaborowski, **P. L. Taylor** et. al. H_0 Without the Sound Horizon (or Supernovae): A 2% Measurement in DESI DR1. arXiv:2411.16677 (2025). (*submitted*).
15. E. Zaborowski, **P. Taylor** et. al. A Sound Horizon-Free Measurement of H_0 in DESI. *Journal of Cosmology and Astroparticle Physics* 2025.06 (2025): 020.
16. Matthew Craigie, **Peter L. Taylor**, Yuan-Sen Ting, Carolina Cuesta-Lazaro, Rossana Ruggeri and Tamara M Davis. Unsupervised Searches for Cosmological Parity Violation: Improving Detection Power with the Neural Field Scattering Transform. *Phys. Rev. D.*, 112(2):023503, 2025.
17. Leah Vazsonyi, **Peter L. Taylor**, Georgios Valogiannis, Nesar S. Ramachandra, Agnès Ferté, and Jason Rhodes. Constraining $f(R)$ Gravity with a k -cut Cosmic Shear Analysis of the Hyper Suprime-Cam First-Year Data. *Phys. Rev. D.*, 104(8):083527, 2021.

²Euclid Consortium Paper.

18. A. Deshpande, **P. L. Taylor**, and T. Kitching. Accessing the high- ℓ frontier under the reduced shear approximation with k -cut cosmic shear. *Phys. Rev. D*, 102(8):083535, 2020.

OTHER PUBLICATIONS

19. Joudaki (... **P. L. Taylor**). Euclid Preparation. Cosmology Likelihood for Observables in Euclid (CLOE). 2. Code Implementation. *A&A Accepted*.
20. Hailey Widger, Andrew Engel, Annika Peter, **Peter L. Taylor**. Spectroscopic Completeness and Photometric Redshift Performance in Astronomical Foundation Models. (Accepted to neurIPS ML4PS, Dec 2025).
21. A. Brodzeller, (... **P. L. Taylor**). Construction of the damped Ly α absorber catalog for DESI DR2 Ly α BAO. *Phys. Rev. D.*, 112:083510, 2025.
22. W. Elbers, (... **P. L. Taylor**). Constraints on Neutrino Physics using DESI DR2 BAO measurements. *Phys. Rev. D*, 15;112(8):083513, 2025.
23. K. Lodha, (... **P. L. Taylor**). Extended Dark Energy Analysis using DESI DR2 BAO measurements. *Phys. Rev. D.*, 112(8):083511, 2025.
24. U. Andrade, (... **P. L. Taylor**). Validation of the DESI DR2 BAO measurements from Galaxies and Quasars. *Phys. Rev. D*. 112(8):083512, 2025.
25. M. Tsedrik, (... **P. L. Taylor 10/11**). Interacting dark energy constraints from the full-shape analyses of BOSS DR12 and DES Year 3 measurements. *Monthly Notices of the Royal Astronomical Society: Letters*, 2025, slaf055.
26. A.G. Adame, (... **P. Taylor**). DESI 2024 VII: Cosmological Constraints from the Full-Shape Modeling of Clustering Measurements. *Journal of Cosmology and Astroparticle Physics* 2025.07 (2025): 028.
27. Sankarshana Srinivasan, Daniel B Thomas and **Peter L. Taylor**. Cosmological gravity on all scales IV: 3x2pt Fisher forecasts for pixelised phenomenological modified gravity. *Journal of Cosmology and Astroparticle Physics*. 2025(02), 071.
28. A.G. Adame, (... **P. Taylor**). DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations. *Journal of Cosmology and Astroparticle Physics* 2025(02), 021.
29. R. Calderon, (... **P. Taylor**). DESI 2024: Reconstructing Dark Energy using Crossing Statistics with DESI DR1 BAO data. *Journal of Cosmology and Astroparticle Physics* 2024.10 (2024): 048.
30. Euclid Collaboration: (... **P. L. Taylor**). Euclid. I. Overview of the Euclid Mission. *A &A special issue 'Euclid on Sky'*. (2024).
31. Kyle Finner (... **Peter L. Taylor 7/8**). Near-IR weak-lensing (NIRWL) measurements in the CANDELS fields I: point-spread function modeling and systematics. *The Astrophysical Journal* 958.1 (2023): 33.
32. A. Ferté (...**Peter L. Taylor 5/6**) et. al. Categorizing models using self-organizing maps: An application to modified gravity theories probed by cosmic shear. *The Open Journal of Astrophysics*, 10.21105/astro.2110.13171, 2023.

33. T. D. Kitching, A. C. Deshpande and **P. L. Taylor**. Spatially varying additive biases in cosmic shear data. *The Open Journal of Astrophysics*, 10.21105/astro.2010.07749, 2021.
34. T. D. Kitching, A. C. Deshpande, and **P. L. Taylor**. Mitigating biases in cosmic shear power spectra amplitude inference. *The Open Journal of Astrophysics*, 10.21105/astro.2110.01275, 2021.
35. A. Deshpande, T. Kitching, V. Cardone, **P. L. Taylor**, S. Casas, S. Camera, C. Carbone, M. Kilbinger, V. Pettorino, Z. Sakr, et al. Euclid: The reduced shear approximation and magnification bias for stage iv cosmic shear experiments. *Astronomy and Astrophysics*, 636, 2020.
36. Thomas D. Kitching, **Peter L. Taylor**, Peter Capak, Daniel Masters, and Henk Hoekstra. Rainbow cosmic shear: Optimization of tomographic bins. *Phys. Rev. D*, 99(6):063536, 2019.
37. Alessio Spurio Mancini, **Peter L. Taylor**, R Reischke, T. Kitching, V. Pettorino, B. M. Schafer, B. Zieser, and P. M. Merkel. 3d cosmic shear: Numerical challenges, 3d lensing random fields generation, and minkowski functionals for cosmological inference. *Phys. Rev. D*, 98(10):103507, 2018.
38. Richard Massey, David Harvey, Jori Liesenborgs, Johan Richard, Stuart Stach, Mark Swinbank, **Peter Taylor** et al. Dark matter dynamics in abell 3827: new data consistent with standard cold dark matter. *Monthly Notices of the Royal Astronomical Society*, 477(1):669677, 2018.
39. M. Jauzac, D. Eckert, J. Schwinn , D. Harvey , C. M. Baugh, A. Robertson, S. Bose, R. Massey (... **Peter Taylor 23/24**) et al. The Extraordinary Amount of Substructure in the Hubble Frontier Fields Cluster Abell 2744, *Monthly Notices of the Royal Astronomical Society*, 463(4), 3876-3893, 2016.

SUBMITTED PAPERS

40. M. Bonici, (... **P. L. Taylor**). cloelike: A Python Library for Cosmological Likelihood Inference in the Euclid Era. arxiv:2605.23841 (2026).
41. M. Bonici, (... **P. L. Taylor**). cloelib: A Flexible Python Library for Computing Cosmological Observables in the Euclid Era. arxiv:2605.23839 (2026).
42. S. Vinciguerra, (... **P. L. Taylor**) . Euclid preparation: Towards a DR1 application of higher-order weak lensing statistics. arXiv:2510.04953 (2025).
43. V. Cardone, (... **P. L. Taylor**). Euclid Preparation. Cosmology Likelihood for Observables in Euclid (CLOE). 1. Theoretical Recipe. arXiv:2510.09118 (2025).
44. G. Canas-Herrera, (... **P. L. Taylor**). Euclid preparation. Cosmology Likelihood for Observables in Euclid (CLOE). 3. Inference and Forecasts. arXiv:2510.09153 (2025).
45. M. Martinelli, (... **P. L. Taylor**). Euclid Preparation. Cosmology Likelihood for Observables in Euclid (CLOE). 4. Validation and Performance. arXiv:2510.09141 (2025).
46. L. Goh, (... **P. L. Taylor**). Euclid preparation. Cosmology Likelihood for Observables in Euclid (CLOE). 5. Impact of systematic uncertainties on the cosmological analysis. arXiv:2510.09147 (2025).

47. L. Blot, (... **P. L. Taylor**). Euclid preparation. Cosmology Likelihood for Observables in Euclid (CLOE). 6. Extensions beyond the standard modelling of theoretical probes and systematic effects. arXiv:2510.10021 (2025).
48. Naim Karacayli and **Peter L. Taylor**. Unified reconstruction of the Lyman-alpha power spectrum with Hamiltonian Monte Carlo. arXiv:2506.08198 (2025).
49. C. Garcia-Quintero, (... **P. L. Taylor**). Cosmological implications of DESI DR2 BAO measurements in light of the latest ACT DR6 CMB data. arXiv:2504.18464 (2025).
50. DESI Collaboration: (... **P. L. Taylor**). Data Release 1 of the Dark Energy Spectroscopic Instrument. arXiv:2503.14745 (2025).
51. DESI Collaboration: (... **P. L. Taylor**). DESI DR2 results I: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints. arXiv:2503.14738 (2025).
52. DESI Collaboration: (... **P. L. Taylor**). DESI DR2 results II: Baryon Acoustic Oscillations from the Lyman Alpha Forest. arXiv: 2503.14739 (2025).
53. L. Casas, (... **P. L. Taylor**). Validation of the DESI DR2 Lya BAO analysis using synthetic datasets. arXiv: 2503.14741 (2025).

NON-REFEREED PAPERS

54. T.D. Kitching, N. Tessore, **P.L. Taylor**. Spatial propagation of weak lensing shear response corrections. arXiv:2302.14656 (2023).